Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[12pt,english,openany,a4paper]book []graphicx []color alltt [T1]fontenc [utf8]inputenc amsmath amssymb

indentfirst

[top=80pt,bottom=60pt,left=78pt,right=78pt]geometry subcaption lipsum booktabs multirow

[english]babel

fancyhdr [C]5

titlesec

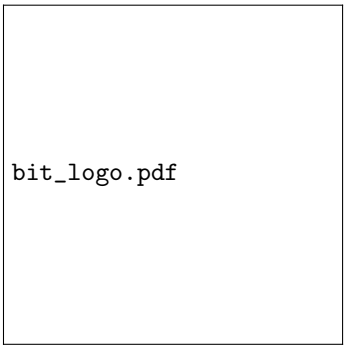
2.1em

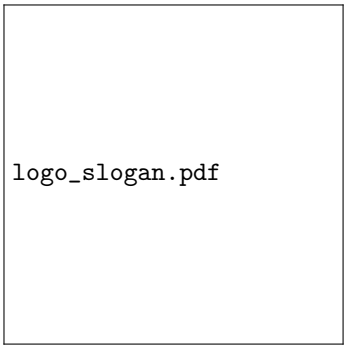
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[backref=page]hyperref [capitalise]cleveref
xcolor
tcolorbox

Course Name

The title of project: A really cool project

Name	James Bond
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logo_slogan.pdf

Abstract

This is a template for Projects/Report/Proposals at BIT. [5]

Keywords: LaTeX, BIT

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Chapter 3

Problem Statement

[1-2]

3.1 Requirements

There are two logos to choose, shown in fig:logo1,fig:logo2.

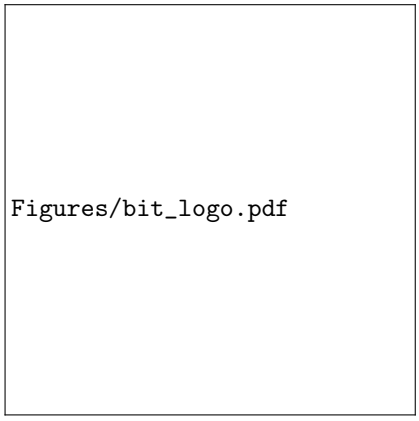


Figure 3.1: Lighter logo

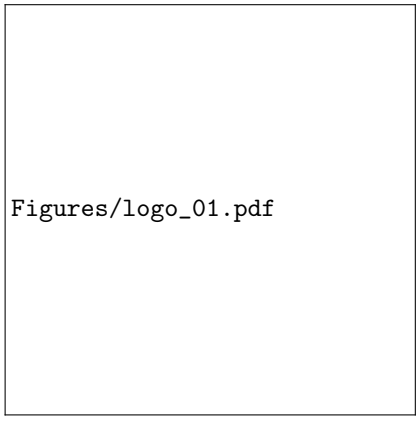


Figure 3.2: Darker logo

The template is designed for customization, in `"Figures/"` folder, `"bit_logo.pdf"` is the left figure and `"logo_01.pdf"` is the right figure.

Chapter 4

Introduction

LaTeX is awesome. [1]

[title=**tcolorbox example**] [2]

Chapter 5

Part Two

Citation is used like[**?**].

0.49

Figure 5.1: Example figure 1

0.49

Figure 5.2: Example figure 2

Figure 5.3: Two figures side by side example 2

fig:figA,fig:figB is an example of two figures side by side.
[6-9]

Table 5.1: The Comparision of Modeling Results

2*Model	Data		
(lr)2-4	Training Set	Test Set	Accuracy
SVM	300	200	84.00%
C4.5	300	200	77.52%

Table 5.2: The Model1 for DoS Attack Data

(lr)2-5 DoS Types	Correct	Wrong	Total	Accuracy
back	0	359	359	0.00%
land	7	0	7	100.00%
neptune	4616	41	4657	99.12%
pod	36	5	41	87.80%
smurf	652	13	665	98.05%
teardrop	12	0	12	100%
mailbomb	0	293	293	0.00%
apache2	72	665	737	9.77%
processtable	135	550	685	19.71%
udpstorm	0	2	2	0.00%

Chapter 6

Conclusions

[2]

Figure 6.1: Our school logo

[3-5]

Appendix

[*Report the config files of the software used (i.e. SU2 and the mesher). Also attach to this report an archive with the mesh files, solutions and the reference solution data (e.g. data points of a Cp plot ...)*]